Bioactive Components in Milk and Dairy Products

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Bioactive Components in Milk and Dairy Products

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Foreword

An old saying declares that "times are changing." This book, *Bioactive Components in Milk and Dairy Products*, is a fine example of the truth in that old saying. For at least the last 100 years dairy science textbooks were about milk production from cows almost exclusively and about cow milk being the basic food for young and old. Now dairy science is joining the globalization of commerce worldwide and opening new focus on the long-neglected diversity of milk production from different mammals and on the complexity of milk composition as a source of a multitude of ingredients. These ingredients provide much more than basic nourishment to young and old; they also offer essential components with significant effects and benefits on body metabolism and body health.

This book provides an important contribution to the new globalized dairy science by detailing what is known of the milk of the other dairy animals whose milk contributes significantly to people's nutrition and health in countries other than the developed West: buffaloes, goats, sheep, camels, and horses. The milk of yaks and reindeer is not covered in this book, but it will be in the future because of the importance of these animals in certain regions of the world.

Bioactive Components in Milk and Dairy Products also presents a comprehensive discussion of the many components in milk of all species; these components have a role beyond the basic nutrients of protein, fat, sugar, and minerals. Chapters detail the chemistry and function of a long list of widely unknown factors that have been proven to have properties and activities beneficial to body health: bioactive milk proteins; β-lactoglobulin; α-lactalbumin; lactoferrin; immunoglobulins; lysozyme; lactoperoxidase; peptides from α_{s1}-, α_{s2}-, and β-; κ-caseins; glycomacropeptides; phosphopeptides; oligosaccharides; conjugated linoleic acid; polar lipids; gangliosides; sphingolipids; medium-chain triglycerides; trans fatty acids; milk minerals; growth factors; and approximately 16 hormones in milk, vitamins, and nucleotides. These proven beneficial effects include being antimicrobial, biostatic, antihypertensive, ACE inhibitive, antiadhesion, antidiabetic, anticholesterol, anticarcinogenic, immunomodulatory, anticariogenic, antiobesity, probiotic, and prebiotic. These factors and effects are discussed for milk and dairy products. Furthermore, regulatory and technological aspects of purification, analyses, and fortification into functional foods are presented.

Renowned world authorities researching the different dairy species and the bioactive components in their milk are the contributors in this book, which makes it especially valuable as a new reference source, for which the editor deserves much credit, and for which a wide distribution of this book is greatly deserved and highly recommended.

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